

The key aspects of security in a database are authentication, authorisation, encryption, firewall and auditing. All these are looked after in MySQL.

Figure 3 shows how MySQL looks at security. This diagram shows us

exactly what we can do in order to assess, prevent, detect and recover any problem that may arise.

MySQL database service on OCI

Everything that you would need to manage MySQL on-premise (like

backup, security patch and upgrade, provisioning and configuring, high availability, etc) is taken care of by the MySQL database service. For example, whenever a new version of MySQL is released, a seamless migration to the newer version is automatically done in a prenotified management upgrade window.

When you are on OCI (Oracle Cloud infrastructure), the latest tools are made available to you. These allow you to write and modify your programs on all the different SDKs, as shown in Figure 4.

MySQL HeatWave

Earlier, if somebody wanted to do analytics, they needed to fetch the data (ETL or extract, transform and load) from the primary database first and then load the analytics database, because often the OLTP and OLAP databases, have differences in features and schemas. That OLAP information then had to be uploaded back into the OLTP database to be queried. This was a major source of time delay and extremely management and manpower and, therefore, cost intensive.

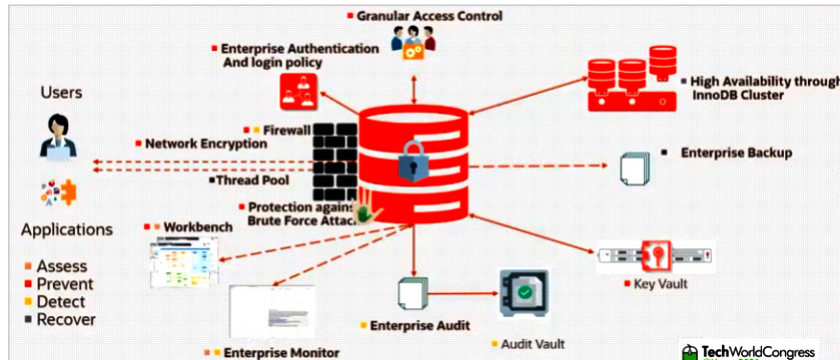


Figure 4: Security architecture of MySQL



Figure 5: Tools and SDKs available on the cloud

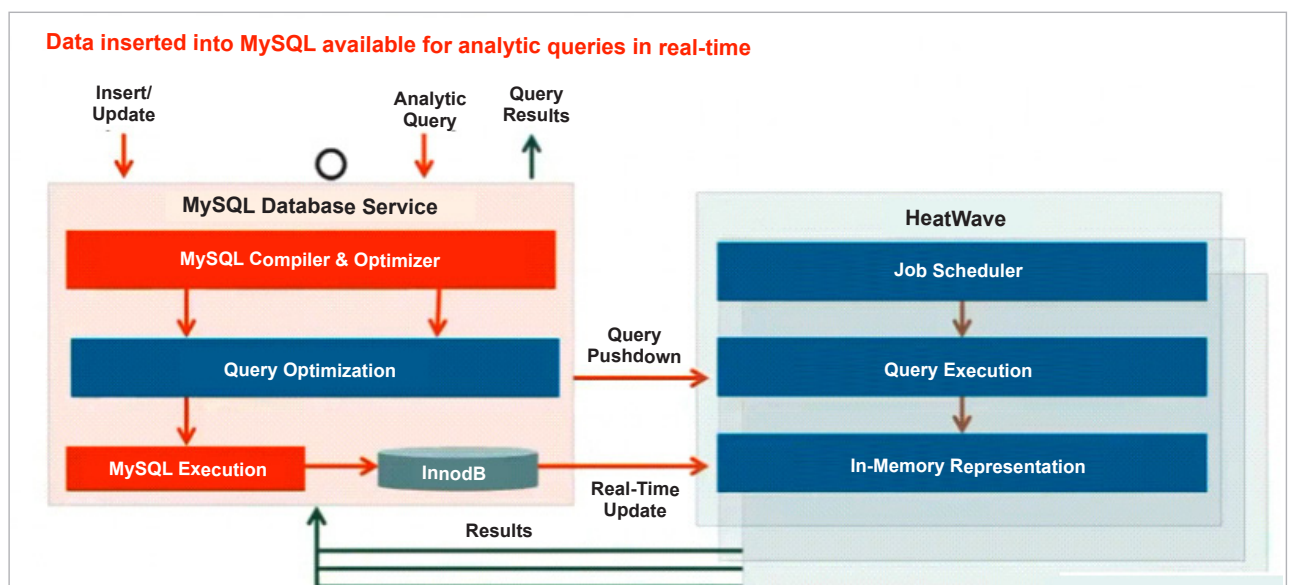


Figure 6: Real-time querying of data in MySQL